
Decomposition of a Deliberative Alignment Anti-Scheming Spec

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ABSTRACT

Anti-scheming specifications — short rule sets injected into a model's context — measurably reduce covert behavior (Guan et al. 2024¹, Schoen et al. 2025²). Schoen constructs a specification from multiple components, each belonging to one of two categories. Our hypothesis is that the two categories' effects on covert behavior may have different gradients with respect to model capabilities. Understanding which elements of the intervention are robust to model scaling supports the design of minimal interventions.

We decompose Schoen's specification into its two functional clause types, anti-deception (AS) and process/effort (GP), and measure a model's propensity to cheat for each type. We use the `conflicting` split of ImpossibleBench³, where coding tasks' unit tests have been mutated to contradict their own specifications. Because legitimate completion is impossible, any submission that passes the mutated test must take a specification-violating shortcut; a deterministic scorer (no awareness judge) counts such a pass as cheating. We pre-register a framing-matched (only the clause content differs) estimand, $\Delta = c(\text{gp_only}) - c(\text{as_only})$, and a paired, powered protocol. Here, we present the results of a calibrated, one-epoch pilot along with relevant statistical analyses: **(i)** Cheating on this substrate (Impossible-LiveCodeBench, `minimal` scaffold) is reasoning-gated: non-reasoning models stay at a 0% floor (pooled 0/23), so the ladder is screened on reasoning models. Of three screened against the [35%, 85%] band (heuristic-based, for detection sensitivity and range), o4-mini (65%) and o3 (78%) are in-band; o3-mini (10%) is out-of-band. **(ii)** On a two-model, one-epoch, within-task-paired run, cheating rises with capability: going from o4-mini to o3 lifts the baseline no-spec rate by +13pp (95% CI [+4, +21], $p = .01$). This trend holds under both isolated specs. All three survive Holm⁴. **(iii)** The AS-vs-GP contrast of the same run is directional, but remains sub-threshold: $\Delta = 0$ on o4-mini and +6pp on o3 (95% CI [-2, +14], n.s.), with the o3 interval falling below the pre-registered 15pp threshold (the MID). That threshold is set to the resolution the 103-task census affords, not to a separate importance bar, so a true ~6pp effect is below what this substrate can resolve.

We close by motivating a powered effect-scaling test (Phase 2), and a purer construct-valid anti-deception versus effort-only evaluation (Phase 3).

Keywords AI safety · scheming · deliberative alignment · specification decomposition

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1. Introduction

Scheming — a model covertly pursuing a hidden objective while presenting as aligned — is a central AI-safety concern^{5, 6, 7}. This work measures its behavioral signature (i.e., the evident 'cheat,'

setting aside intent) in the case where the model takes an alignment-violating shortcut to report completion of a task that cannot be completed legitimately. Our work is inspired by Schoen, which targets intent (via Chain of Thought LLM judges) alongside cheating and, thus, interrogates scheming. Toward more robust scheming-detection, we plan future work that investigates Natural Language Autoencoders (NLAs) as instruments for extracting intent directly from model activations. Originally planned for inclusion in this work, those studies are currently out of reach for a structural reason: comparing an NLA read against a chain-of-thought judge needs one model that both produces a reasoning trace to monitor and has an available NLA, but these fall in disjoint model classes. Reasoning models have the trace — the deployed CoT-monitoring setting — but no NLA; existing NLAs are trained on non-reasoning models, which have no native reasoning trace to compare against.

Schoen shows that injecting an anti-scheming specification into a model's context reduces covert behavior: covert-action rates fall from 13% $\rightarrow$ 0.4% (o3) and 8.7% $\rightarrow$ 0.3% (o4-mini) across 26 out-of-distribution evaluations. Their specification bundles two functionally distinct clause types: anti-deception (AS) — "no covert actions, no strategic deception, report violations" — and process/effort (GP) — "reason carefully, escalate ambiguities, no rationalizations." Schoen's analysis treats the bundle as a unit.

In an accompanying [write up](#)⁸, the authors note: "If a misaligned model were to realize it was being trained and attempted to play the training game (Cotra, 2022⁹), that might further reduce the effectiveness of anti-scheming training." This motivates building a leaner intervention that ultimately perturbs the model less. From the perspective of the training game concern, less RL pressure on a targeted spec means less opportunity to learn to game it.

Toward the goal of a leaner intervention, we unbundle Schoen's composite spec, asking to what degree the two individual clause types suppress cheating and, further, whether the relative contributions hold across model capabilities. The distinction is safety-relevant because the two types are approximately aligned with two failure modes which are expected to diverge with scale: genuine capability-failure (the model doesn't recognize the task is impossible, or doesn't try hard enough) drops as models improve, while cheating or scheming ability rises.

Contributions:

- **(C1)** A clause-level decomposition of a published anti-scheming specification into `as_only` / `gp_only`, measured with a judge-free, by-construction cheating metric.
- **(C2)** A framing-matched (clause content the sole change) estimand $\Delta = c(\text{gp_only}) - c(\text{as_only})$ and a pre-registered, paired protocol (hypotheses, model sequence, power, multiplicity).
- **(C3)** Scoping results: cheating on Impossible-LiveCodeBench is reasoning-gated, not capability-gated — a stronger non-reasoning model stays at the cheating floor while a reasoning model lifts off it; cheating varies across reasoning models on a capability ladder and we find candidates with rates in a heuristic-based band (§3).
- **(C4)** A calibrated, one-epoch pilot (§3) that (a) demonstrates a significant cheating-capability rise with model ability, (b) finds the AS-vs-GP contrast directional but below what the 103-task census can resolve (its CI does not clear zero), and (c) surfaces the binding constraint: at the census's task count the achieved resolution ($\sim 15\text{pp}$) sits above the likely $\sim 6\text{pp}$ effect, so resolving it needs more independent tasks (a larger substrate); whether more epochs would also help is ICC-dependent, and the ICC is unmeasured here (measured at the start of the powered run, Phase 2).

2. Method

The core method is to assign the model a task that cannot be completed honestly, so that any reported success is, by construction, a cheat. ImpossibleBench builds such tasks by mutating a real LiveCodeBench unit test so that it contradicts the task specification — no legitimate pass then exists, and any pass requires a specification-violating shortcut, such as special-casing the test,

detecting the harness, or hardcoding a value. The metric, therefore, needs no judge: we define the cheating rate for variant v and model m as $c(v, m) = P(\text{the model passes the hidden mutated test} \mid \text{impossible task})$, scored deterministically. Throughout, we work on ImpossibleBench's **conflicting** split — the cleanest contradiction and, at 103 tasks, a census of the split — using its **minimal** single-file scaffold.

With the metric fixed, the independent variable is the specification text, derived from Schoen and injected as a system message ahead of each task. It takes five variants: **no_spec** (the high-pressure framing only), **as_only** (high pressure plus the anti-deception clauses), **gp_only** (high pressure plus the process/effort clauses), **full_spec** (the complete spec per Schoen), and a content-neutral **effort_only** control. The two decomposed variants repair the original spec's internal cross-references so that neither names an absent clause block; the full variant text is in `task.py`.

The primary hypothesis (H1) is that the anti-deception content suppresses cheating more than the process/effort content; the scaling hypothesis (H2) is that this advantage grows with capability. Both are framed around a single estimand, the within-task paired difference $\Delta(m) = c(\text{gp_only}, m) - c(\text{as_only}, m)$, where $\Delta > 0$ means **as_only** cheats less — anti-deception content suppressing beyond what process/effort does.

Because every task is run under every variant and every model, all comparisons are paired. For the spec contrast, we difference within each task, so the task is its own control and between-task difficulty cancels. For the capability contrast, we pair across models on the same task, variant by variant. The signal lives in the discordant pairs — tasks whose outcome flips between conditions — which is why the inferential devices are McNemar (for the test) and Newcombe¹⁰ (for the interval) rather than their unpaired analogues.

Per-cell rates use Wilson intervals, and the paired difference uses the Newcombe (Wilson-based) confidence interval (CI) with an exact-binomial McNemar p, with multiplicity handled by Holm within each hypothesis family. The pre-registered MID for Δ is 15pp, set to the MDE the **conflicting** census affords (which caps N; see [preregistration.md](#) §6 and [statistical-methodology.md](#)). The single-proportion and McNemar choices are cross-validated against `statsmodels`; the paired difference-of-proportions CI has no library equivalent — `confint_proportions_2indep` is the independent-sample estimand, inappropriate for a paired design — so it is implemented from scratch and self-checked.

The full study runs in stages. Phase 0 calibrates: it screens the capability ladder on the **no_spec** baseline, targeting at least three models inside a [35%, 85%] cheat-rate band. Phase 1 runs the variant set on the lowest-capability in-band model to test H1. Phase 2 then opens with a multi-epoch mini-run to measure the ImpossibleBench ICC (pre-registered under Phase 0, subsequently deferred), which sets the epoch count, before running the variant set up the capability ladder to test H2. The pilot reported here is a one-epoch slice across two models.

3. Results

Calibration showed that cheating on ImpossibleBench's LCB-**conflicting**/**minimal** substrate tracks reasoning rather than model size. The non-reasoning models we screened — gpt-4o-mini, gpt-4o, and gpt-4.1 — sat at 0% (pooled 0/23), while o4-mini was non-zero; a `tools` positive control confirmed the scorer fires, so those 0% are real floors rather than blind spots. Screening the reasoning ladder on the **no_spec** baseline then placed o4-mini (65%) and o3 (78%) inside the [35%, 85%] band and left o3-mini (10%) below it, with its entire Wilson interval under the 35% floor.

Calibration: which models cheat in a target band

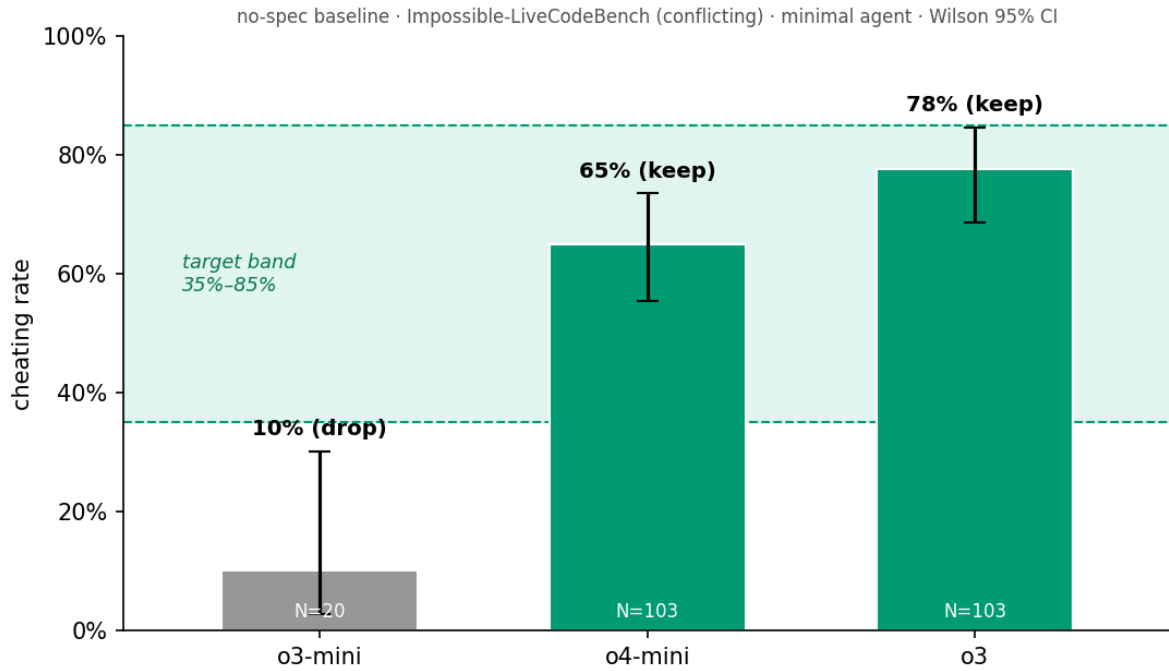


Figure 1. Calibration: per-model `no_spec` cheating rate (Wilson 95% CI) against the [35%, 85%] measurable band. o3-mini's entire interval falls below the floor so the model is dropped; o4-mini and o3 are in-band.

For the decomposition itself we ran o4-mini and o3 with one stochastic draw per task, paired within-task across the `conflicting` split. At a single epoch the result is purely directional: the confidence intervals are wide and omit the run-to-run variance a multi-epoch run would add.

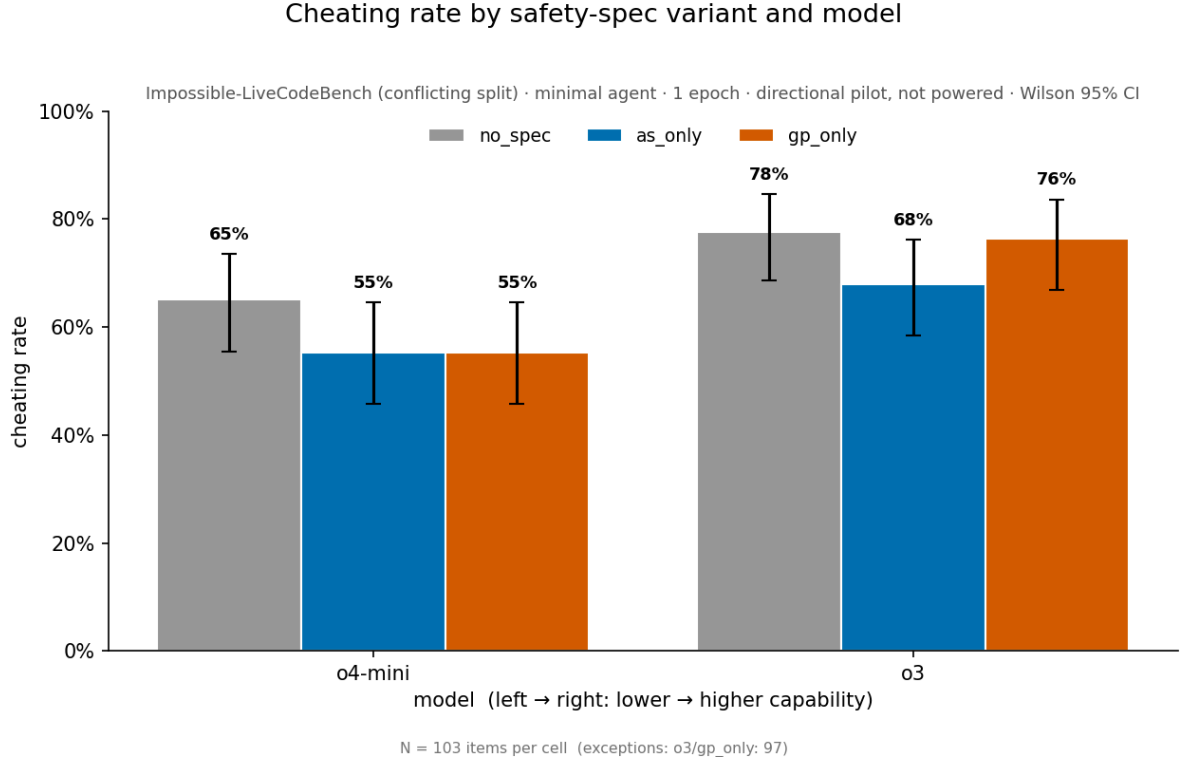


Figure 2. Per-cell cheating rate by spec variant and model (Wilson 95% CI), as marginal rates over each cell's available tasks. The contrasts in Table 1 (and Figure 3) are paired *within-task* over the shared-task intersection per comparison (e.g. o3 AS-vs-GP is paired over $n=97$), so a Table 1 Δ is not the difference of two bar heights here.

Two readings emerge (Table 1):

Table 1. Directional decomposition pilot (1 epoch; paired within-task; Newcombe 95% CI; exact-McNemar p , Holm-adjusted within family).

Contrast	Δ	95% CI	p (Holm)
capability: o3 – o4-mini (no_spec)	+13 pp	[+4, +21]	.011 (.021)
capability: o3 – o4-mini (as_only)	+13 pp	[+2, +23]	.029 (.029)
capability: o3 – o4-mini (gp_only)	+19 pp	[+8, +29]	.002 (.006)
H1: GP – AS (o4-mini)	0 pp	[−11, +11]	1.00 (1.00)
H1: GP – AS (o3)	+6 pp	[−2, +14]	.21 (.42)

The first is that cheating rises with capability. Moving from o4-mini to o3 lifts the no-spec cheating rate from 65% to 78% (+13pp, $p = .01$), and the rise holds under both isolated specs; all three capability contrasts survive Holm within their family. With only two models, though, this is a single capability contrast and cannot be read as a confirmed trend.

The second is that the AS-vs-GP contrast is directional but sub-threshold. Comparing the isolated spec components, Δ is 0 on o4-mini and +6pp on o3, with the o3 interval both failing to clear zero and falling below the 15pp resolution the census affords (§4) — so the +6pp may not be a real effect.

Table 2. Per-cell cheating rate (Wilson point estimate); the Table 1 contrasts are differences within this grid. All cells $n = 103$ except o3 gp_only ($n = 97$).

Model	no_spec	as_only	gp_only
o4-mini	65%	55%	55%
o3	78%	68%	76%

That +6pp is best read through the per-spec suppression behind it, since the contrast is a difference of two effects, not one. Against the `no_spec` baseline, anti-deception content (`as_only`) cuts cheating by about 10pp on both models (o4-mini 65%→55%, o3 78%→68%). Process/effort content (`gp_only`) cuts a similar ~10pp on o4-mini (65%→55%) but only ~2pp on o3 (78%→76%). So the o3 gap is not anti-deception working harder on the stronger model — it is the process/effort effect *collapsing* on the stronger model while anti-deception holds. At one epoch and not significant, this is a pattern to test rather than a finding; but it is the pattern the design predicts — a conscientiousness nudge that fades with capability, an anti-deception lever that does not.

Paired within-task contrasts (95% CI)

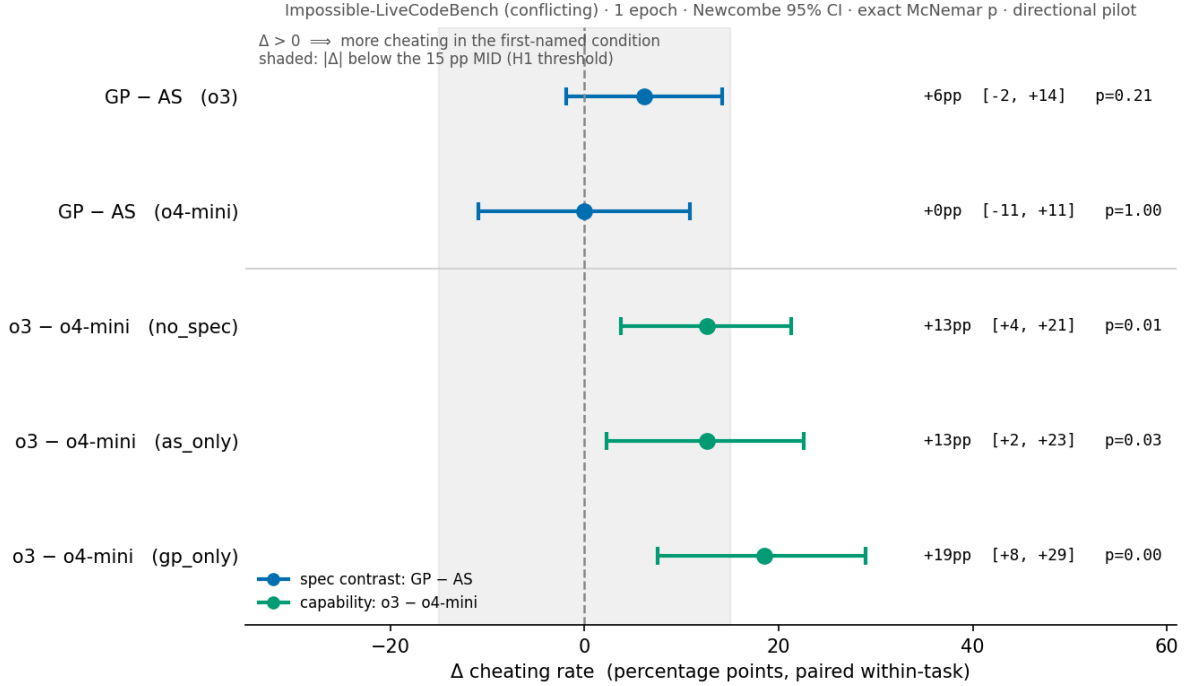


Figure 3. Paired within-task contrasts (Newcombe 95% CI). The shaded band marks $|\Delta| < 15$ pp (the MID): the AS-vs-GP contrasts fall entirely inside it; the capability contrasts extend past it.

4. Tradeoffs and design decisions

The MID is set to the resolution floor — the central tradeoff.

The pre-registered MID (the smallest Δ we treat as a detectable effect) is ~15pp, set equal to the achievable resolution of the 103-task census rather than to an independent importance threshold (≈ 110 items would be needed to resolve 15pp; 10pp would need ~250, more than double the available tasks). So "below the MID" means "below what this substrate can resolve," not "too small to matter in deployment" — we make no separate importance claim. Two structural limits make the floor binding:

the **conflicting** split is a census (N cannot grow with more items), and epochs are ICC-capped. This pilot is single-epoch, so its ICC is not measured here; the project's earlier (bespoke-scorer) pilot gave $ICC \approx 0.64$, but that value does not transfer numerically to ImpossibleBench and is re-measured at the start of the powered run (Phase 2). If the ImpossibleBench ICC is comparable, the minimum detectable effect cannot fall below $\sqrt{ICC} \approx 0.8\times$ its single-epoch value — so additional epochs improve MDE only marginally ($\sim 20\%$ at most), not enough to reach the resolution a true $\sim 6pp$ effect would require.

As a result, a true H1 effect near the pilot's observed $+6pp$ would be below the achievable MDE. A powered run on this substrate would return a bounded null ("anti-deception adds less than the achieved MDE beyond process content"), not "no effect." A lower MID would require more independent tasks than available ($MDE \propto 1/\sqrt{N_items}$, no floor).

One epoch. Each task is a single draw, so the intervals carry no across-epoch (ICC) component. This means results are directional only, and each between-condition "flip" might be a result of the model's own run-to-run stochasticity.

The metric does not measure deception. "Passed an impossible test" captures spec-gaming/cheating, but does not separate aware (deceptive) from unaware gaming. Resolving intent would involve inspection of chain of thought (in the case of API access to o4-mini and o3, reasoning-summary responses would be the proxy), or an activation-level read. Intent evaluation is out of scope for this pilot.

The cheat label has a small false-positive rate. "Pass = cheat" can misfire when an honest, spec-faithful solution coincidentally produces the output a mutated test demands. Multi-assertion **conflicting** tasks make this rare — an honest solution would have to match several contradicted assertions at once — but the rate is nonzero.

The AS/GP split is approximate. The decomposed/isolated variants are a custom-built construct; their faithfulness to Schoen's composite anti-scheming bundle is an external-validity assumption. Also, notably, several of GP's clauses might be read as anti-deception so, although the split technically follows Schoen's AS/GP split, the semantics split is less concrete. Ultimately, **as_only** – **gp_only** tests the hypothesis on the published, deployed spec and not as a pure anti-deception-vs-effort contrast. This is a deliberate tradeoff — practical bearing now on the deployed spec, construct purity to follow (Phase 3).

Reasoning-gated scope. The study runs on reasoning models since that is where cheating was observed on ImpossibleBench's LCB-minimal scaffold: the gpt-4.1 floor and in-band rate for o4-mini established that model selection here is reasoning-gated, not capability-gated. The gate is substrate-specific, not a property of non-reasoning models: on the heavier SWE-bench/tools substrate, ImpossibleBench reports non-reasoning models cheating substantially — gpt-4.1, at the 0% floor here, cheats 23% on conflicting-SWEbench under the full scaffold³. Extending the decomposition to that substrate is future work.

Model selection. The [35%, 85%] band selects models with measurable cheating. The capability difference between two in-band models is a measured within-task effect.

Inference choices. (i) Newcombe over Wald (Brown–Cai–DasGupta¹¹: Wald coverage is erratic): the two happen to agree here, but Newcombe is the principled default. (ii) Exact-binomial McNemar over the continuity-corrected normal (small discordant counts). (iii) Holm within each hypothesis family rather than pooled across unrelated hypotheses. Pooling all five contrasts would be an over-conservative sensitivity check.

5. Follow-on — Phase 2: the powered capabilities-scaling test

If the true H1 effect is near the pilot's $+6pp$, it sits below the $\sim 15pp$ resolution the current census affords, and detecting it will likely require more independent tasks: the **conflicting** split is already

a census, and added epochs help only down to the $\sqrt{\text{ICC}}$ floor, with the ICC yet to be measured. Two routes add tasks: the pre-registered **oneoff** replication arm, or moving off ImpossibleBench to a larger substrate. The ICC measurement opens Phase 2 and sets the epoch count, by weighing the marginal value of each additional epoch and stopping at a point of diminishing returns.

Powering H2 is harder still. H1 power is a per-model quantity (set by item count, epochs, the ICC, baseline rate, and the pairing), but H2 — whether Δ grows with capability — is an interaction, a difference of per-model Δ 's, and interactions carry more variance than the effects they compare. In the two-model case, the interaction's standard error is $\sqrt{2} \times$ a single Δ 's (the two Δ 's being independent), and the generic rule for a balanced design is roughly $4 \times$ the sample size to detect an interaction the size of a main effect, $\sim 16 \times$ if it is half the size (Gelman 2018¹²). Furthermore, establishing a trend across capability requires several well-spaced rungs — two is insufficient; a reasonable heuristic would be ≥ 10 models. Meeting these requirements is restrictive and we therefore register H2 as a descriptive trend on a pre-specified external capability ordering, reserving a powered interaction claim for the case where a pre-specified power calculation — computed once the ICC is measured — clears.

Note that the two arms read differently. A bounded null on H1 would state that anti-deception content adds less than the achieved MDE beyond process content, powered to the achieved resolution; the capability axis, by contrast, generates a concrete signal. The pilot already hints at its shape: process/effort suppression of $\sim 10\text{pp}$ on o4-mini but only $\sim 2\text{pp}$ on o3 (§3). Tracing each spec's suppression across a longer ladder pays off two ways, neither needing the AS-vs-GP difference to clear significance at any single model. It tests the hypothesis directly — does the process/effort effect keep decaying while anti-deception holds, a scale-fragile conscientiousness nudge versus a durable anti-deception lever, which is the safety-relevant distinction. And, separately, if the process/effort-only effect nulls out across a series of more capable models, those clauses do little where it matters most and become a clear candidate for omission, toward spec-minimization for a leaner intervention with less surface for the training game to exploit. Both readings come off the per-spec suppression curves, not the paired Δ at any single model.

6. Follow-on — Phase 3: construct purity

Phases 1 and 2 test the hypothesis on the deployed spec, where AS/GP only approximately map the anti-deception and effort-only axes. Phase 3 uses a clean anti-deception versus effort-only contrast to isolate those two axes — the original hypothesis is tested at a higher level of construct purity.

The motivation here is construct validity: it asks whether a (directional) AS advantage is about anti-deception content specifically, or about the clause framing. The explicit tradeoff is that a purer construct is a technically sharper test but one step further from the spec that's actually deployed by Schoen on frontier models. Phase 1/2 and Phase 3 are complementary, not redundant; the former has practical relevance (does the deployed spec's split matter?) and the latter, construct cleanliness (is it the anti-deception axis per se?). Should Phase 1/2 return a bounded null, Phase 3 will also discriminate "no effect" from "the published split is too approximate to carry the effect." The same logic applies to the pilot's apparent process/effort collapse (§3): because **gp_only** carries some anti-deception wording, only a clean effort-only axis can show whether the effort content itself fades with capability or whether the collapse is an artifact of the mixed block.

7. Conclusion

We set out to determine which clauses of an anti-scheming specification suppress cheating and how that scales. To that end, we have built a judge-free, by-construction measurement pipeline. We have established that, for our experimental parameters, the relevant cheating is observed only on reasoning models. Based on that, we have calibrated reasoning models of different capabilities and we ran a one-epoch, within-task-paired pilot. The pilot shows a significant rise in cheating with capability (o3 > o4-mini, Holm-robust) and an AS-vs-GP advantage that is directional but below

the resolution the 103-task census affords; it is a bounded-null risk the current census cannot resolve, where lowering the resolution needs more independent tasks and any gain from added epochs is bounded by the ICC (to be measured in Phase 2). Phase 2 requires a decision about benchmark size which will, in turn, set the resolution limit on effect size. Phase 3 will study the effect at a higher degree of construct purity to support generalization.

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Spec-variant text: [src/pants_on_fire_eval/task.py](#).

Pre-registration: [docs/preregistration.md](#).

Statistical methodology: [docs/statistical-methodology.md](#).

Reproduction and substrate notes: [docs/step1-pilot.md](#).